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AHA SCIENTIFIC STATEMENT

Early Recognition and Intervention for Poststroke Spasticity: A Scientific Statement From the American Heart Association

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ABSTRACT: Spasticity and related motor disorders are common and often disabling complications after stroke, affecting an estimated 30% to 80% of survivors. Spasticity can impair functional mobility, reduce independence, and increase caregiver burden. Secondary complications, including pain, restricted range of motion, skin breakdown, and joint contractures, further degrade quality of life, limit rehabilitation outcomes, and increase health care costs. Despite the availability of options to manage spasticity and mitigate its effects, timely diagnosis and intervention remain key challenges. Many patients receive treatment only after spasticity has become well established, and others receive no treatment at all due to persistent disparities in recognition, access, and delivery of treatment for spasticity, contributing to long-term disability and increased costs of care. This scientific statement reviews the rationale, established and emerging evidence, and strategic approaches for improving early recognition and intervention for poststroke spasticity. Early intervention is defined as treatment initiated within the first 3 months after stroke onset, potentially before development of secondary impairment from poorly controlled spasticity. Recognizing spasticity as a multidomain clinical syndrome—including involuntary muscle overactivity, impaired voluntary motor control, and passive tissue remodeling—offers important opportunities to improve timely diagnosis and treatment. Progress will also depend on a deeper understanding of the time course and pathophysiology of spasticity through both animal and human models. This scientific statement also outlines strategies to close gaps in recognition and care, including expanding and training the specialist workforce; developing innovative, scalable approaches for early detection and management; and strengthening care pathways and access to meet the substantial unmet needs of patients with poststroke spasticity.

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Spasticity is a common and often debilitating consequence of stroke, affecting an estimated 25% to 80% of survivors.[1–6] This translates to ≈1.8 to 5.6 million stroke survivors in the United States alone. Globally, spasticity is estimated to affect >100 million individuals after stroke.[6] Poststroke spasticity substantially affects functional recovery, quality of life, and long-term independence.[5]

The 2016 American Heart Association/American Stroke Association guidelines for adult stroke rehabilitation and recovery note that the cost of care is 4 times higher when spasticity is present and suggests approaches for recognition and treatment of spasticity.[7] Nevertheless, spasticity remains underrecognized and undertreated. For many patients, this results in prolonged disability and the emergence of preventable secondary complications, which increases overall health care costs.[2]

The optimal timing for initiation of spasticity-related interventions is a crucial yet unresolved aspect of poststroke care. For the purposes of this scientific statement, early intervention refers to treatment initiated within the first 3 months after stroke onset. The 3-month time window is consistent with previous studies of early spasticity

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intervention,[8,9] and is not meant to exclude treatment approaches that might be initiated earlier in the poststroke course.

This scientific statement provides an overview of the current landscape of the diagnosis and management of poststroke spasticity, reviews the rationale for and available evidence supporting early treatment, and outlines future opportunities to reduce spasticity through early rehabilitative approaches. We identify key strategies to improve timely spasticity care by increasing awareness among patients and health care professionals to facilitate early recognition of risks and symptoms, and developing cost-effective, scalable models to improve access to care, especially for high-risk individuals. Addressing these challenges will be essential for improving outcomes for the millions of people with poststroke spasticity.

METHODS

The American Heart Association/American Stroke Association Stroke Rehabilitation and Recovery Committee convened a diverse group of experts in stroke research and clinical care, with specialty expertise that included neurology, physiatry, neurosurgery, physical therapy, and neuroscience research. A stroke survivor provided personal experience highlighting the importance of early treatment, emphasized many of the opportunities to improve care, and reviewed the overall manuscript. For this scientific statement, we searched PubMed for English-language publications from January 2010 through March 2025. In reporting treatment effects (Table), we prioritized conclusions from randomized controlled clinical trials (>40 participants) and meta-analyses. Additional studies were included if higher-quality evidence was not available or if newer therapeutic interventions were discussed.

CURRENT UNDERSTANDING OF AND APPROACHES FOR POSTSTROKE SPASTICITY

Definitions and Clinical Manifestations of Spasticity

Spasticity is classically defined as a "velocity-dependent increase in tonic stretch reflexes with exaggerated tendon jerks, resulting from hyperexcitability of the stretch reflex, as one component of the upper motor neuron syndrome."[10] This is assessed clinically by passively stretching each muscle group at slow and fast speeds; in the presence of spasticity, faster stretches elicit reflexive contraction (the "spastic catch"), leading to increased resistance to movement. In practice, the spectrum of deficits extends to long-term patterns of intermittent and sustained muscle overactivation, and is not confined to abnormal responses to passive muscle movement. Stroke-related injury involving cortical motor neurons and their descending axonal tracts spans 3 interrelated domains (Figure 1) that together degrade functional mobility: (1) abnormal involuntary muscle activity, (2) impaired voluntary motor control, and (3) passive tissue remodeling. Incorporating a more comprehensive view of spasticity enhances insight into the patient experience, integrates advances in pathophysiology, and creates opportunities for new approaches to diagnosis and treatment.

In addition to overactive motor reflexes, other forms of involuntary muscle activity associated with spasticity include clonus (rhythmic contractions triggered by muscle stretch), involuntary spasms, and postural hypertonia—also referred to as spastic dystonia—characterized by sustained tonic contraction in the absence of voluntary movement.[11] These manifestations typically involve the same muscle groups affected by spasticity (upper limb: shoulder adductors, elbow, wrist, and finger flexors[11]; lower limb: knee extensors and ankle plantar flexors[12,13]) and are generally managed with similar therapeutic strategies.

Impaired voluntary motor control is a defining feature of poststroke paresis. Weakness often progresses in a proximal-to-distal gradient[14–16] and frequently coexists with spasticity, resulting in spastic paresis. Loss of selective motor control presents as difficulty activating muscles at a single joint independently. Patients also often struggle to coordinate proximal and distal segments of the arm or leg independently, resulting in stereotyped movement patterns known as flexion and extension synergies. These abnormal synergies manifest in an effort-dependent manner and can severely limit function.[15,17–19]

Patients with spasticity also develop secondary structural (passive) tissue changes that contribute to resistance during both passive and active movement. These include muscle stiffness, shortening, and eventually fibrosis of muscle and joint capsules, which can lead to fixed joint contractures.[20,21] Distinguishing increased resistance due to structural changes from reflex-mediated spasticity can be challenging without advanced tools, such as electromyography to detect myoelectric activity during stretch[22] or ultrasound to identify fibrotic changes.[22]

Together, involuntary muscle overactivity, abnormal voluntary control, and passive tissue remodeling form a self-reinforcing triad. Sustained abnormal limb postures due to postural hypertonia, disruption of purposeful movement by abnormal voluntary control, pain, and the learned nonuse of the affected limb accelerate structural changes, worsening range of motion and impairing effective contraction. The time course of development of muscle overactivity and abnormal voluntary control varies

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Table. Summary of Evidence-Based Management Categories for Poststroke Spasticity

Primary target and treatment category	Level of evidence	Clinical action
Involuntary muscle activation		
Botulinum toxin (BoNT-A)	A*	Consistently reduces MAS; early use with optimized dosing and ultrasound guidance enhances outcomes
Intrathecal baclofen	A	Effective for generalized spasticity; requires monitoring for adverse effects
Selective dorsal rhizotomy	A	Effective in lower extremity spasticity; used selectively
Transcutaneous electrical nerve stimulation	A	Effective in lower limbs; duration of effect and upper limb efficacy remain unclear
Phenol or alcohol nerve blocks	B*	Provide temporary tone reduction; option when BoNT-A is unavailable or ineffective
Oral antispasticity agents (eg, baclofen, tizanidine)	B/C	Modest tone reduction; often limited by sedation or systemic side effects
Selective motor neurectomy	C	Invasive option with longer-lasting effects; limited availability
Cryoneurolysis	Emerging	Emerging evidence for sustained tone reduction; more research needed
Abnormal voluntary control		
Task-specific training	A*	Improves motor control and function
Constraint-induced movement therapy	A	Promotes use of affected limb; best suited for higher-functioning patients
Functional electrical stimulation	A* (LE), B (UE)	Enhances voluntary muscle activation; optimal results depend on timing
Noninvasive brain stimulation (rTMS/tDCS)	A	Effective in upper limbs; strongest support for anodal tDCS and low-frequency rTMS
Repetitive peripheral magnetic stimulation	A	Reduces MAS; treatment measures require further refinement
Bimanual training	B*	Supports interlimb coordination; more robust trials needed
Music-supported therapy	B/C*	Encouraging early evidence; supports motor function through auditory-motor coupling
Robotic therapy	B	Facilitates repetitive, task-specific training; promising in combination with conventional therapy
Vagal nerve stimulation	B	Improves Fugl-Meyer upper extremity scores when paired with therapy
Aerobic exercise and endurance training	C*	Improves motor control and counters effects of inactivity
Passive tissue changes		
Early mobilization and weight-bearing interventions	A*	Helps prevent contractures and reduce hyperreflexia when started early
Passive stretching exercises	A	Not effective as standalone treatment; must be combined with other rehabilitation modalities
Static stretching with positioning orthoses	A	Modest MAS improvements; evidence limited by small studies
Dry needling	A	Short-term benefit in lower limbs; effects diminish by 4 wk
Extracorporeal shockwave therapy	A	Short-term MAS reduction; radial may be superior to focused therapy
Whole-body vibration	A	More effective in younger patients; strongest support for LE with <20 Hz, 10-min protocols
Localized muscle vibration	A	Improves MAS in elbow or wrist; limited data on long-term and lower limb effects
Serial casting	B	Temporarily prevents contractures; requires labor-intensive application
Soft tissue interventions (eg, massage, myofascial release)	C	Limited evidence; frequently used as adjunct to other therapies
Hyaluronidase	Emerging	Improves MAS and ROM in subacute or chronic spasticity; emerging therapeutic role

This brief summary does not divide tested therapies according to timing or function (eg, upper extremities [UEs] or lower extremities [LEs]). MAS indicates Modified Ashworth Scale; ROM, range of motion; rTMS, repetitive transcranial magnetic stimulation; and tDCS, transcranial direct current stimulation.

*Evidence of effectiveness when given in early stroke (<3 months from onset) based on Tilborg et al.[8]

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on the basis of the severity of both motor and sensory impairment, and occurs earlier—often within the first month—in individuals with more severe impairment.[23,24] Resulting stiffness and shortening of spastic muscles follows, and worsens over the first 6 months to a year, further disadvantaging the already weak muscles by requiring greater force to overcome resistance.[25] Over time, this cycle entrenches abnormal patterns, disuse, and functional limitations.[26] Improved understanding of these components through research and clinical practice is essential to guide timely, targeted interventions and optimize outcomes.[2]

Assessment and Diagnosis of Spasticity

Accurate diagnosis and quantification of spasticity is imperative for effective management and evaluation of treatment outcomes. Current tools for diagnosing and assessing spasticity include clinical rating scales, electrophysiologic methods, and biomechanical instruments.

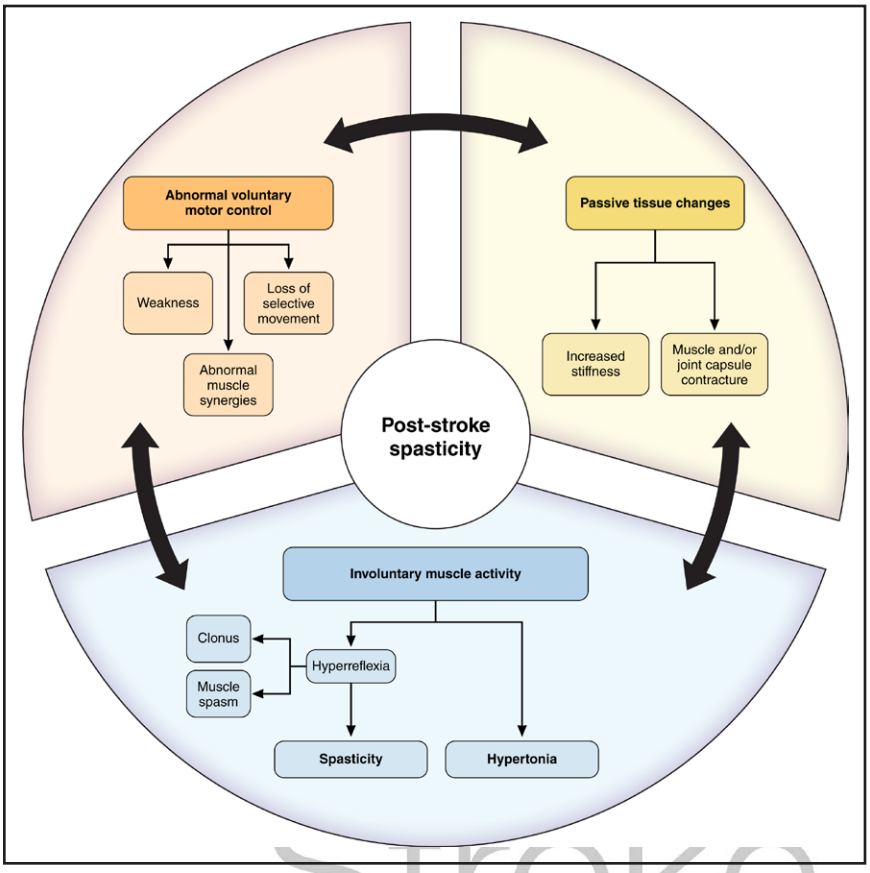

Figure 1. Spectrum of motor dysfunction in poststroke spasticity. Spasticity is narrowly defined as velocity-dependent hyperactivity after rapid muscle movement. The full syndrome of interacting deficits in patients with stroke includes impaired voluntary muscle control, involuntary muscle overactivation at rest (hypertonia) or with movement (hyperreflexia), and passive tissue remodeling.

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Each offers useful information but has limitations in capturing the full complexity of spasticity and its functional impact. The most widely used clinical rating tools are the Modified Ashworth Scale (MAS) and the Modified Tardieu Scale. Both scales assess resistance to passive movement, but have limited ability to distinguish between resistance caused by increased involuntary muscle activity versus passive tissue changes.[27] The MAS provides a rapid, subjective estimate of muscle tone by grading resistance during slow passive movement. A score of ≥1 is commonly used to indicate spasticity. However, the MAS does not account for the velocity-dependent nature of spasticity, and may overestimate tone in the presence of joint stiffness or contracture. The Modified Tardieu Scale improves upon the MAS by incorporating passive movements at different velocities and measuring the angle at which the catch occurs, indicating reflex onset, offering a more precise assessment of stretch reflex hyperexcitability.[28,29] However, the Modified Tardieu Scale is more time-consuming and less frequently implemented in routine clinical settings.

Although clinical scales remain foundational for both research and practice, they require in-person assessment by trained examiners and are subject to interrater variability. Functional assessments, such as evaluations of gait or upper limb task performance using standardized activities, can help contextualize findings.[30] A recent scientific statement supports a 5-step framework for assessment of spasticity that includes evaluation of passive movement, active movement, fatigability during repeated movement, and upper and lower limb function.[30,31] Measurement approaches that integrate a broader assessment of spasticity may more accurately reflect real-world functional effects of spasticity on mobility, independence, and quality of life.

Pathophysiology of Poststroke Spasticity

Effective interventions for spasticity rely on a comprehensive understanding of its underlying pathophysiology. The initial driver of stroke-induced motor deficits, including spasticity, is interruption of the descending corticospinal tract (Figure 2), which is the primary conduit for voluntary movement in humans and is uniquely responsible for generating dexterous, precise, and complex actions. Consequently, motor weakness (ie, paresis) is typically the earliest clinical manifestation of corticospinal injury in acute stroke. In contrast, the hallmark feature of spasticity—hyperactivity of the muscle stretch reflex—emerges days to weeks after stroke onset.[32] This reflects abnormal facilitation of the monosynaptic intraspinal circuit linking muscle spindles and Golgi tendon organs, which sense muscle length and stretch, through 1a and 1b afferent fibers to α motor neurons, which produce muscle contraction.[33] The extent of muscle activation associated with postural hypertonia directly influences spasticity

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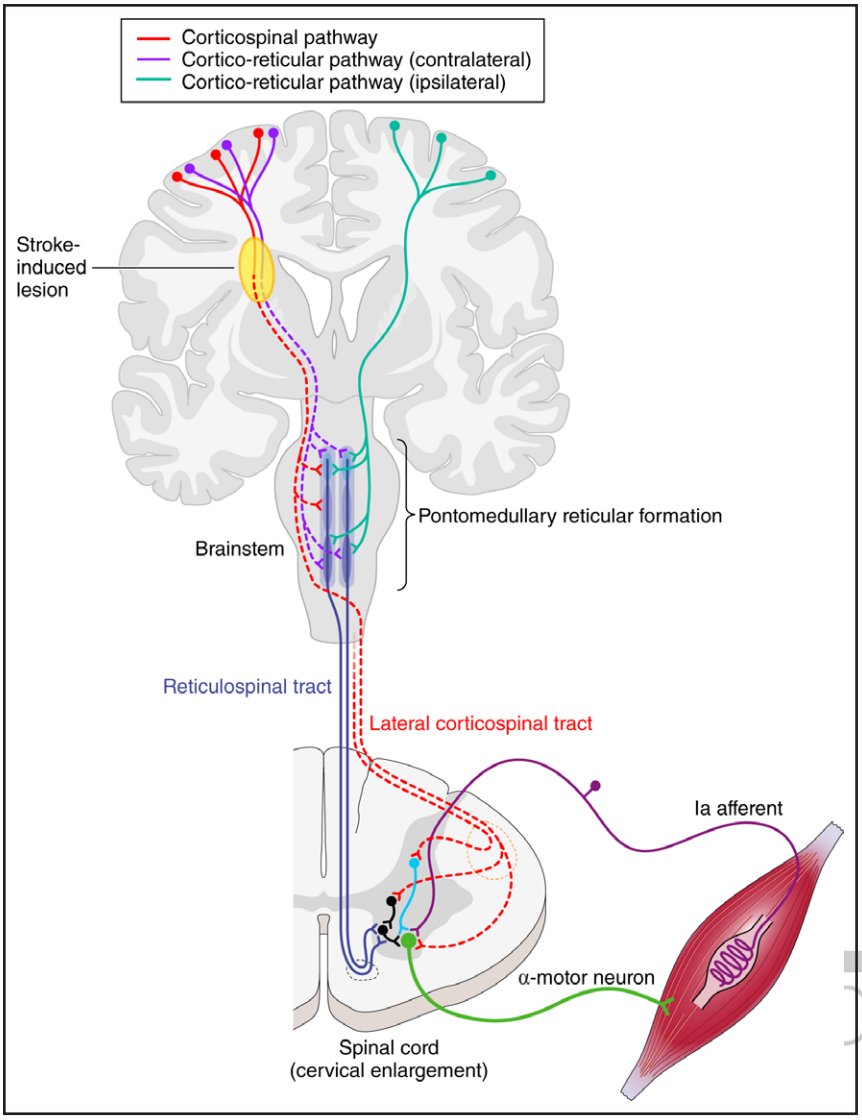

Figure 2. Proposed mechanisms of supraspinal involvement in poststroke plasticity.
Schematic shows some of the descending motor control pathways and their potential interaction with intraspinal circuits. The corticospinal tract (red) controls fine motor activity by direct activation of spinal motor neurons (green) as well as indirect activation through excitatory and inhibitory interneurons. Extrapyramidal tracts control muscle tone and posture through an indirect pathway that involves motor nuclei in the brainstem. The schematic shows the corticoreticular tract (purple), with axons derived from sensory and motor cortex, which synapse on brainstem reticular nuclei, which in turn send reticulospinal axons to spinal neurons. At each spinal cord level, intraspinal circuits include 1a and 1b (not shown) afferents, spinal interneurons, and α motor neurons. Stroke-induced interruption of the corticospinal tract causes acute weakness and subsequent spasticity on the opposite side of the face and body. The features of spasticity reflect abnormal hyperactivity in intraspinal reflex arcs and overactivation of descending pathways, such as the reticulospinal tract.

severity.[34,35] Pathologic hyperactivity within these spinal reflex circuits can occur from interruption of corticospinal fibers alone.[36,37]

Accumulating evidence indicates that in patients with stroke, muscular hyperactivity is likely enhanced by pathologic activation of brainstem supraspinal pathways that normally modulate motor tone and reflex excitability.[38–40] The various supraspinal pathways—including the vestibulospinal, rubrospinal, cerebellar spinal, and reticulospinal tracts—have different contributions to spontaneous and stroke-induced motor activity depending on species and experimental methods. In humans, the reticulospinal tract has received considerable attention, both for its role in normal movement[41–43] and its contribution to poststroke motor impairment.[19,41,44–46] Therapeutic progress requires a deeper understanding of the specific supraspinal and intraspinal neural pathways, cellular targets, and neurotransmitters leading to dysregulation of motor tone and reactivity after stroke.

In addition to these neural mechanisms, spasticity can promote secondary, non-neural alterations in muscle and connective tissue, which further limit movement and exacerbate spasticity. These changes include shortened muscle fascicle length and joint or limb stiffness caused by increased accumulation of extracellular matrix fibrotic components, such as hyaluronan and collagen.[20] Elevated hyaluronan content disrupts the viscoelastic and lubricating properties of muscle tissue, increasing friction and impairing fiber gliding during movement.[21,26] Over time, hyaluronan aggregation may induce fibroblast-to-myofibroblast transformation, leading to irreversible muscle fibrosis and joint contractures.[47] Such biomechanical alterations can occur independently of neural hyperexcitability and may persist even after neuromodulatory treatments (eg, botulinum toxin), underscoring the contribution of passive tissue stiffness to spasticity.[48]

There has been considerable progress in identifying pathophysiologic mechanisms underlying poststroke spasticity, but clinical translation has been hindered,

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in part, by limitations of commonly used animal models. Spasticity is difficult to quantify reliably in rodents, and there are key differences in rodent, nonhuman primate, and human neural pathways relevant to motor control and recovery.[49,50] Further research is needed to translate mechanistic insights into effective therapeutic approaches for human poststroke spasticity.

Current Treatment Approaches

The American Heart Association/American Stroke Association guideline for poststroke rehabilitation includes the following recommendations: (a) "management of spasticity may include stretching, splinting, oral medications, and chemodenervation (eg, botulinum toxin injections)" and "for patients with stroke and focal spasticity interfering with function or care, botulinum toxin may be considered."[7] Whereas the most studied treatment approaches for spasticity have been primarily geared toward reducing involuntary muscle activity, newer approaches targeting abnormal voluntary motor control and passive tissue changes are under investigation and may warrant inclusion in future clinical practice guidelines. The Table provides a list of available treatment approaches for poststroke plasticity, along with proposed mechanisms of action and the level of evidence supporting treatment efficacy.

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RATIONALE AND EVIDENCE SUPPORTING EARLY TREATMENT

Effective management of poststroke spasticity is a vital component of stroke recovery and rehabilitation. Although multiple therapeutic modalities have demonstrated efficacy (Table), optimal timing and selection of interventions remain areas of active investigation. Delayed or omitted treatment in patients with established, symptomatic spasticity can lead to substantial harm, including increased pain, progressive motor dysfunction, and secondary musculoskeletal complications, such as stiffness, muscle shortening, and joint contractures.[51–53] These sequelae may further restrict mobility, limit the effectiveness of rehabilitation, and impose greater burden on caregivers.

Delays in spasticity treatment are often attributable to underrecognition of early symptoms, lack of health care professional training, and fragmented referral pathways, leading to missed opportunities for timely intervention and complication prevention.[5,54] Limited access to rehabilitation services and reimbursement barriers exacerbate management delays and increase all-cause health care costs for people with spasticity.[2] Emerging mechanistic insights and recent studies increasingly support the rationale for earlier intervention—before impairment is fully established. Early treatment may promote favorable outcomes for stroke survivors by preserving functional mobility, supporting reparative plasticity, improving treatment responses, and reducing secondary musculoskeletal complications.

Early Spasticity Treatment Improves Mobility During the Critical Window for Recovery

The early poststroke period is characterized by a heightened capacity for neuroplasticity, often referred to as the critical window for motor recovery. This period, typically within the first 3 to 6 months, is when intensive, task-specific training is most effective in promoting motor relearning, functional gains, and both behavioral compensation and true recovery.[55,56] However, spasticity can emerge within the first few weeks after stroke onset,[57,58] interfering with voluntary movement and limiting a patient's ability to engage in rehabilitation. Pain associated with spasticity further disrupts sleep and reduces participation in key motor training activities.[33,59–61] Untreated spasticity during this sensitive period may undermine the overall effectiveness of neurorehabilitation and contribute to persistent long-term disability. Early intervention aimed at reducing spasticity during this phase of maximal spontaneous plasticity may help optimize the benefits of neurorehabilitation and improve long-term outcomes.

Early Activity May Promote Reparative Plasticity and Reduce Immobility and Abnormal Postures That Lead to Chronic Musculoskeletal Disability With Secondary Effects on Spasticity

The relationship between paresis and spasticity is complex and dynamic. Severe weakness and postural hypertonia lead to prolonged immobility and sustained abnormal postures, which result in adaptive changes within the musculotendinous unit, such as loss of sarcomeres and decreased muscle fiber fascicle length in muscles affected by spasticity,[62,63] and increased stiffness in muscles held in a specific position for prolonged periods,[11] which results in a loss of range of motion[64]; exacerbates spasticity, pain, and fatigue[65]; and may lead to contractures.[26] Preventing or minimizing these structural maladaptations through early mobility and other interventions may reduce long-term disability and improve functional outcomes.

The close interactions of paresis and spasticity must be recognized: targeting spasticity alone may not improve functional capacity if underlying weakness is not also addressed.[33] The temporal relationship between paresis and spasticity underscores the need for tailored, adaptive interventions that address the evolving neurophysiologic state of the recovering nervous system. Timing and intensity of early mobilization remain important

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variables; in addition, incorporating task-specific mobility interventions during the subacute phase may be a promising strategy for reducing the severity or incidence of spasticity.[66] We review the evidence supporting early intervention in this context.

Evidence for Early Treatment of Established Spasticity

A recent systematic review evaluated the effectiveness of early interventions for poststroke spasticity.[8] Drawing from 27 studies, the review assessed physical, pharmacologic, and neuromodulatory treatments using the International Classification of Functioning framework. The most robust evidence supported early administration of botulinum toxin. Ten studies demonstrated that botulinum toxin significantly reduced involuntary muscle activation in the short term, measured primarily by the MAS or electromyography. Some trials also reported functional gains, such as improvements on the Modified Barthel Index, although these were less consistently observed across the activity and participation domains.[67] Although larger trials are needed, these studies demonstrate clinically relevant benefits even when botulinum toxin is administered early in the course of spasticity development.

Functional electrical stimulation and task-specific training also showed probable benefit, with 4 of 6 included studies reporting significant reductions in spasticity when applied early.[8,68] However, whether the effects are related to motor recovery or reduction in spasticity is unclear. The use of transcutaneous electrical nerve stimulation, transcranial direct current stimulation, and robot-assisted therapy was associated with mixed or inconclusive results, largely due to methodologic variability and small sample sizes.[8]

Interventions targeting passive muscle changes, such as orthoses and shockwave therapy, also showed mixed results. One study specifically examined the early use of shockwave therapy and found a beneficial effect in preventing the progression of spasticity.[69] Further research is warranted to validate and expand upon these initial results.

Only 2 studies directly compared early versus late intervention. An observational, routine practice study involving 303 patients found that MAS score reduction was greater in the early treatment start group (start time, 3.74±1.75 months [mean±SD]) but was not statistically different from the medium treatment start (start time, 20.11±11.08 months) or late treatment start (start time, 144.24±90.85 months) groups at the end of the study.[59] A similar longitudinal cohort study of 83 patients found that MAS scores were significantly lower at 4 and 12 weeks in patients who first received botulinum treatment <3 months after stroke onset compared with those who received treatment >3 months after stroke.[60] Randomized controlled and dosing studies are needed to examine long-term outcomes of early versus late intervention using botulinum toxin.

It is plausible that early mobilization and interventions that reduce involuntary muscle activity early after stroke may promote better rehabilitation engagement and reduce the likelihood of passive changes in muscles and long-term disability. However, high-quality randomized and long-term studies are needed to confirm whether early intervention leads to durable improvements in functional independence. Despite promising findings, there is no universally accepted critical window for spasticity intervention as exists for motor recovery.

CAN SPASTICITY BE PREVENTED?

Although spasticity is a common and often debilitating consequence of stroke, it is not an inevitable outcome. Identifying individuals at highest risk may offer an opportunity for early intervention.[70] Patients with initial features such as severe motor paresis, early hyperreflexia, or lesions involving the internal capsule or brainstem are particularly vulnerable. In these individuals, proactive strategies may help prevent the onset or reduce the severity of chronic spasticity.

Randomized controlled trials on primary prevention of spasticity are limited, but observational and cohort studies support the concept of risk reduction rather than absolute prevention. Early interdisciplinary management guided by clinical risk assessment tools may reduce the incidence or progression of spasticity, preserving functional independence and reducing the burden of long-term care. Preventive measures include early passive mobilization,[65,68,71] active rehabilitation using approaches such as robotic-assisted therapy to provide proximal joint stabilization to target the flexor synergy pattern,[72,73] and use of stretching devices.[74] Pharmacologic treatment initiated within the first 3 months after injury may further reduce the risk of persistent spasticity and its long-term functional consequences.[70]

Mobilization in the first days after stroke has been associated with multiple benefits, including a potential role in the prevention of spasticity. Initiating mobilization within 24 to 72 hours after stroke, as part of comprehensive rehabilitation, may help preserve muscle length, reduce neural hyperexcitability, and maintain joint range of motion, thereby mitigating the risk of spasticity.[75] Combining botulinum toxin injections with early motor training appears to be the most beneficial strategy for modulating muscle tone and improving motor recovery, compared with delaying treatment or using these treatments separately.[76] Overall, although more targeted trials are needed, early mobilization remains a promising strategy for spasticity prevention and reduction within integrated stroke care.

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STRATEGIES TO IMPROVE EARLY RECOGNITION AND TREATMENT

Persistent gaps in awareness, assessment, and clinical implementation contribute to the widespread underutilization of effective therapies for poststroke spasticity, leading to preventable disability and suboptimal outcomes for many stroke survivors. Spasticity often goes unrecognized in its early stages and, even when identified, treatment is frequently delayed or omitted altogether.[2] Management is complex, and many clinicians—particularly those without specialized training—struggle to determine appropriate interventions. Access to experienced clinicians is also limited, especially in rural and underresourced areas. In the United States, for example, the ratio of Medicare beneficiaries to clinicians administering botulinum toxin exceeds 50 000:1, even in large metropolitan regions.[4] These gaps underscore the need for a more proactive and coordinated approach to early spasticity management. Key strategies to address early recognition and treatment are summarized in Figure 3 and detailed in the following sections.

Strengthen Spasticity Risk Assessment and Diagnosis

Timely risk assessment is essential to enable early intervention and reduce the disabling consequences of poststroke spasticity. Multiple clinical and neuroanatomic predictors of spasticity have been consistently identified. Among the most reliable are the severity of initial sensorimotor impairment and early onset of spasticity during the acute poststroke phase. Hemorrhagic stroke and ischemic strokes affecting the anterior circulation—especially those involving the corticospinal tract, internal capsule, or basal ganglia—are associated with elevated spasticity risk. Additional risk factors include stroke lesions >3 cm in diameter and an National Institutes of Health Stroke Scale score ≥10. The PREP2 score (Predict Recovery Potential 2)[77] is an algorithmic system to predict upper extremity function at the 3-month mark but does not include measures of spasticity.

The MAS with a threshold of ≥1 is the most frequently used clinical metric for diagnosing spasticity in research and clinical settings. However, the MAS lacks sensitivity, and may fail to detect early or subclinical spasticity.[78] These observations suggest the need for validated, standardized clinical screening approaches that can be used for stepwise assessment by nonspecialists in primary care and inpatient settings.

Emerging technologies offer the potential for more objective and reproducible assessments. These include biomechanical sensors to quantify movement resistance or joint stiffness; automated motion analysis systems to provide detailed kinematic data; physiologic instruments, such as surface electromyography, H-reflex amplitude, and reciprocal inhibition studies; and imaging-based predictors to identify lesion patterns associated with spasticity.[79] Although not yet part of routine clinical practice, these tools are increasingly used in research settings and clinical trials. They offer valuable insights into the distinct neural and biomechanical components of spasticity and may ultimately facilitate personalized, mechanism-informed interventions.

Risk stratification tools validated predictive models, and electronic medical record analysis[2,80] could support individualized monitoring protocols and early therapeutic planning. Future efforts should focus on integrating clinical, neurophysiologic, and functional data to enable more accurate risk profiling and personalized treatment pathways.

Establish Techniques for Remote Assessment

Validated methods for remote spasticity assessment have the potential to facilitate earlier diagnosis and improve access to care, especially for patients in medically underserved or geographically remote areas. Video-based assessments can allow clinicians to remotely evaluate range of motion, gait abnormalities, and task performances, offering qualitative insights into the presence and functional impact of spasticity. However, the clinical definition of spasticity—characterized by velocity-dependent resistance to passive movement—traditionally requires hands-on testing, limiting its applicability in telehealth settings. The lack of standardized assessment tools that do not require in-person physical examination remains a barrier. Verduzco-Gutierrez and colleagues have proposed a framework for spasticity evaluation by telemedicine, which includes assistance of a caregiver in demonstration of range of motion for video assessment.[31] In addition, patient- and caregiver-reported outcome measures such as ArmA and LegA[81,82] may offer tools for subjective assessment. Caregiver burden may be assessed using the Zarit Burden Interview.[83] Emerging technologies present promising alternatives. Wearable technologies, including surface electromyography and inertial measurement units, can capture abnormal muscle activity and deviations in joint kinematics.[84,85] Data from such devices may be securely transmitted to clinicians, supporting dynamic, data-informed adjustments to treatment plans. These tools provide objective, real-time biomarkers that may facilitate early recognition of spasticity and enable continuous at-home monitoring during the subacute recovery phase.[86] Further research is needed to refine, validate, and standardize remote assessment protocols across diverse patient populations and care environments, which may require integration of patient-reported outcome measures, video analysis, and wearable sensor data.

Establish Opportunities for Remote Therapy

Telehealth platforms increasingly support patients with neurologic disorders by providing care beyond specialized care centers.[87] Telemedicine and virtual consultations offer scalable solutions for extending specialty

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	Early Recognition and Diagnosis	Early Treatment
Medical and Technology Development	• Develop and validate risk prediction scores for spasticity • Identify clinical, imaging, and molecular biomarkers for prediction, monitoring, and treatment response • Create tools for remote assessment and objective quantification (eg, wearable sensors, EMG, imaging)	• Advance basic and translational research seeking new treatment approaches based on pathophysiology of disrupted motor circuits • Develop scalable, cost-effective technologies for spasticity management • Develop remote treatment platforms (eg, tele-rehabilitation, digital therapeutics)
Clinical Trials	• Assess the impact of early recognition or risk stratification on treatment decisions and outcomes	• Conduct trials comparing early (<3 months) vs delayed intervention • Include validated spasticity outcomes in post-stroke studies • Evaluate cost-effectiveness and quality-of-life benefits of early treatment
Implementation	• Integrate screening tools and risk scores into acute and post-acute stroke care pathways	• Embed evidence-based spasticity management protocols in standard care pathways • Develop clinical algorithms for primary care to support early recognition and management of spasticity, pain, and musculoskeletal complications
Education	• Provide accessible educational resources for patients and caregivers on early signs of spasticity • Develop and disseminate educational materials for early recognition of spasticity for health professional	• Create targeted training programs for healthcare professionals on early identification and intervention • Establish certification programs in spasticity management for multidisciplinary teams, including primary care
Access to Care	• Expand telehealth and mobile outreach for early assessment, especially in underserved areas • Streamline referral pathways from stroke and primary care teams to rehabilitation specialists	• Increase the workforce trained in spasticity care (eg, physiatrists, neurologists, therapists) • Address reimbursement barriers for timely interventions and medications in both inpatient and outpatient settings

Figure 3. Potential strategies to improve early recognition and treatment of poststroke spasticity.

expertise to rural or resource-limited settings, enabling timely assessment and treatment planning when in-person care is not feasible.[88] Telehealth approaches have been tailored to support remote spasticity management in patients receiving botulinum toxin.[88] Standard in-person therapy protocols can be modified to provide effective physical, occupational, and speech therapy, and increase access to patients who are homebound or far from treatment facilities.[89] Beyond synchronous televisits, digital platforms offer tools such as exercise libraries, wearable sensor feedback, and patient-reported outcome tracking, potentially empowering patients and caregivers to maintain daily therapy routines, enhancing adherence, and facilitating early detection of changes.

To realize the full potential of remote spasticity therapy, future priorities include developing standardized protocols and reimbursement frameworks to ensure equitable access and sustainable models, validating sensor-based biomarkers against in-person measures of tone and function, and designing integrated telehealth pathways that blend synchronous therapy, remote monitoring, and automated feedback within coordinated care plans.

Expand Patient and Health Care Team Awareness

Expanding awareness of poststroke spasticity among patients, health care professionals, and payors is

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essential to improving early recognition, timely intervention, and long-term functional outcomes. Although spasticity affects up to 80% of stroke survivors, it remains underdiagnosed and undertreated. Many patients do not recognize symptoms such as muscle tightness or involuntary contractions as medical concerns, and health care professionals—especially those outside of neurorehabilitation—may underestimate the impact of spasticity on mobility, pain, and quality of life.[90]

Active engagement strategies that emphasize building trustful, long-term relationships, codesigning interventions with stakeholders, and adapting approaches to fit local contexts have been shown to be helpful in improving processes that affect quality of care.[91] Success depends on open communication, shared decision-making, and continuous feedback loops across all levels of implementation. Empowering patients and caregivers through training for earlier recognition and reporting of symptoms may prompt evaluation before contractures or irreversible tissue changes develop. Storytelling and patient narratives are powerful, culturally relevant tools for raising public awareness, humanizing the condition, and mobilizing support from legislators, health system leaders, and insurers.[92] These narratives demonstrate the tangible benefits of treatment and reinforce the message that spasticity is a treatable complication, and early intervention can prevent long-term disability. Clinician education is equally important. Many primary care clinicians, therapists, and general practitioners lack familiarity with the signs of spasticity or the full range of effective interventions. Professional training should emphasize screening tools such as the Poststroke Checklist and promote timely referral to spasticity specialists.[93] Broader uptake may require the use of implementation science methods and tools to systematically address barriers to implementation of best practices for screening and treatment of spasticity.[94] A national consensus survey highlighted the lack of standardized clinical pathways and called for coordinated, multidisciplinary care models supported by clear referral protocols and systemic awareness efforts.[90] Addressing these widespread educational gaps is a vital step toward reducing the long-term disability burden of poststroke spasticity and ensuring equitable access to timely, evidence-based care.

Expand Access to Spasticity Care

Improving access to spasticity care requires addressing system-level barriers that delay diagnosis and limit treatment, especially among underserved populations. Key strategies include expanding the trained workforce, integrating spasticity care into multidisciplinary care models, and reforming reimbursement mechanisms to support early and sustained intervention.

Expanding clinical expertise in spasticity management will require targeted investment in training at multiple levels. This includes specialized fellowships, interdisciplinary residencies, continuing medical education, and structured mentorship programs. Optimal spasticity management requires coordination of multidisciplinary care teams.[95] Education should also be integrated into the core curricula for physical and occupational therapists, rehabilitation nurses, stroke nursing navigators and coordinators, nurse practitioners, and physician assistants to ensure that frontline clinicians can recognize early signs and initiate appropriate referrals.

Coordinated, multidisciplinary care models are essential for improving outcomes and ensuring continuity of care. Standardized referral pathways and electronic health record prompts can streamline patient identification and accelerate specialist involvement. In parallel, care navigation services can support patients and families by clarifying treatment options, resolving logistical barriers, and promoting adherence to therapy plans. These team-based approaches have been shown to improve functional outcomes, reduce caregiver burden, and enhance health system efficiency.[96]

Cost remains a substantial barrier to long-term spasticity care, particularly for ongoing treatments such as botulinum toxin injections. Stroke-related spasticity is associated with increased health care utilization, including higher rates of hospitalization, outpatient visits, and disability-related interventions.[97] Pharmacologic treatment accounts for only a small fraction of these costs. Emerging evidence suggests that early intervention may be cost-saving: a secondary cost-consequence analysis of a randomized controlled trial found that early administration of botulinum toxin significantly reduced contracture treatment costs without increasing overall costs of care.[98,99]

Policy engagement is also essential. Early spasticity care should be positioned as a core component of postacute stroke rehabilitation, with support for inclusion in bundled payment models and long-term care planning. Education campaigns can encourage broader coverage of underutilized, evidence-based treatments. Policy strategies—such as mandating spasticity screening at hospital discharge or early outpatient follow-up—may improve functional outcomes, reduce health care utilization, and limit disability progression.

Enhancing access to spasticity care through workforce development, telehealth expansion, care coordination, and reimbursement reform represents a key opportunity to improve outcomes and reduce the long-term burden of stroke-related disability.

CONCLUSIONS AND FUTURE DIRECTIONS

Reframing poststroke spasticity as a potentially preventable and treatable complication—rather than an inevitable consequence—offers an opportunity to reduce the global burden of stroke-related disability. Achieving this goal requires a coordinated, multidisciplinary effort to target all 3 contributors to spasticity: involuntary muscle activation, impaired voluntary motor control, and secondary passive

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tissue changes. Early recognition and intervention in each of these domains can improve functional recovery, minimize long-term disability, and reduce overall health care costs. Key strategic considerations arising from this scientific statement are summarized in Figure 3.

Poststroke spasticity is a chronic condition that may emerge in the acute phase and persist for months or years, leading to progressive functional decline. A systematic, multimodal treatment strategy incorporating rehabilitation, pharmacologic, and neuromodulatory approaches should be initiated early and tailored to each patient's evolving needs. Such an approach has the potential to alter long-term trajectories and improve quality of life for stroke survivors.

The increasing prevalence of stroke survivorship has amplified the socioeconomic burden of spasticity. However, access to specialized care remains limited, particularly in rural and underserved communities.[100] Addressing this gap will require expansion of the trained clinical workforce, integration of spasticity care into coordinated stroke pathways, and adoption of remote assessment and treatment technologies to extend reach beyond traditional care settings.

Raising awareness among patients, clinicians, and payors is equally vital. Improved education and outreach can facilitate earlier recognition, timely referral, and broader implementation of evidence-based interventions. Promoting early spasticity care as an essential component of poststroke rehabilitation will be key to driving systemic change.

Despite persisting uncertainties around timing, patient selection, and long-term cost-effectiveness, early diagnosis and intervention for poststroke spasticity represent a promising opportunity to enhance recovery and reduce disability. Future research should focus on validating these strategies in real-world settings and defining optimal care models that can be implemented equitably across diverse health care systems. Advancing these priorities will help transform the current care landscape and improve outcomes for millions of stroke survivors worldwide.

ARTICLE INFORMATION

The American Heart Association makes every effort to avoid any actual or potential conflicts of interest that may arise as a result of an outside relationship or a personal, professional, or business interest of a member of the writing panel. Specifically, all members of the writing group are required to complete and submit a Disclosure Questionnaire showing all such relationships that might be perceived as real or potential conflicts of interest.

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*Modest.

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*Modest.

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